Financial Engineering Final End-Semester Comprehensive Report

# Capital Budgeting, Infrastructure Project Valuation & Clean Energy Portfolio Optimizer


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| **Data Sources:** Kaggle / World Bank PPI Database, SECI & MNRE Records | **Platform:** Live Streamlit Cloud Web Application (PyTorch GPU-Ready) |


## 1. Executive Motivation & Project Scope (Final End-Semester Defense)

Large-scale clean infrastructure projects—such as a **100MW Solar Park in Rajasthan (■250 Crores)** or a 50MW Wind Farm in Tamil Nadu (■350 Crores)—involve substantial Day-0 capital outlays funded by 70% bank loans. Traditional single-scenario Excel sheets fail to capture multi-decade revenue risks, bank debt service coverage, and portfolio diversification. This project provides a full-fledged quantitative decision suite delivering: **(1) Deterministic Multi-Stage DCF & LCOE**, **(2) Bank Loan Debt DSCR Amortization**, **(3) ESG Carbon Offset Monetization**, **(4) 5,000-Scenario Monte Carlo Tail-Risk Engine**, and **(5) Multi-Asset Markowitz Portfolio Optimizer**.

## 2. Mathematical Formulations Across the 4 Core Analyses

### Analysis 1: Deterministic Multi-Stage DCF, NPV, IRR & LCOE

- **Net Cash Flow ($CF_t$) = ($Revenue_t$ - $OpEx_t$) × (1 - Tax Rate)**
- **Net Present Value (NPV) = -CapEx + $\sum_{t=1}^{N}$ [ $CF_t$ / $(1 + WACC)^t$ ] (NPV = +■275.50 Cr)**
- **Internal Rate of Return (IRR): Root where NPV(IRR) = 0 solved via Brent's method (IRR = 18.52% vs WACC = 8.00%)**
- **Levelized Cost of Energy (LCOE in ■/kWh) = [ Total Discounted Lifetime Costs (■) ] / [ Total Discounted Energy (kWh) ] = ■1.99/kWh**
- **Profitability Index (PI) = Present Value of Inflows / CapEx = 2.10x | Discounted Payback = 7 Years**

### Analysis 2: Project Finance Debt Structuring & ESG Carbon Offsets

- **Bank Loan Debt Service Coverage Ratio ($DSCR_t$) = $CFADS_t$ / [ Principal $Amortization_t$ + $Interest_t$ ]**
- **Banking Decision Rule: Project is Bankable if Minimum DSCR ≥ 1.30x (Our Project: Min DSCR = 1.54x, Average = 1.82x)**
- **Annual Carbon Abatement = Annual Generation (MWh) × 0.82 $tCO_2$/MWh = 147,600 Tonnes $CO_2$ avoided/yr**
- **Carbon Revenue Stream = 147,600 tonnes × ■1,000/tonne = +■14.76 Crores/year extra income (+21.3% revenue expansion)**

### Analysis 3: Stochastic Monte Carlo Simulation (5,000 Iterations)

- **3 Joint Random Shocks: Tariff ~ N(μ = ■3.85, σ = 15%), CapEx ~ U(1.0, 1.25), Inflation ~ N(μ = 2.5%, σ = 1.0%)**
- **Quantitative Tail-Risk: Probability of Loss = 0.00% | 95% Value-at-Risk (VaR) = +■152.80 Crores**

### Analysis 4: Multi-Asset Markowitz Clean Energy Portfolio Optimizer

- **Portfolio Return: $R_p = \sum w_i \times IRR_i$ | Portfolio Variance: $\sigma_p^2 = w^T \Sigma w$**
- **Correlation: Solar (Day) vs Wind (Night) = -0.25 (Drastically reduces portfolio cash flow variance from 14.5% to 6.85%)**
- **Optimal Allocation (■1,000 Cr Fund): 45% Solar + 25% Wind + 20% Hybrid BESS + 10% Hydro -> Sharpe = 1.48**

## 3. Clean Infrastructure Benchmark Dataset (12 Representative Assets)

| Project Name | Sector | CapEx (■ Cr) | Gen (MWh) | Tariff (■/u) | LCOE (■/u) | NPV (■ Cr) | IRR (%) | Decision |
|---|---|---|---|---|---|---|---|---|
| Rajasthan 100MW Solar | Solar | ■ 250 | 180k | ■ 3.85 | ■ 1.99 | **+■ 275.5** | **18.5%** | ■ ACCEPT |
| Tamil Nadu 50MW Wind | Wind | ■ 350 | 130k | ■ 4.40 | ■ 4.45 | **-■ 74.9** | **5.5%** | ■ REJECT |
| Gujarat Desalination | Water | ■ 320 | 95k | ■ 5.80 | ■ 4.75 | **+■ 68.2** | **10.2%** | ■ ACCEPT |
| Kutch 150MW Hybrid+BESS | Hybrid | ■ 520 | 310k | ■ 4.60 | ■ 3.10 | **+■ 312.4** | **15.8%** | ■ ACCEPT |
| Tehri 200MW Pumped Hydro | Hydro | ■ 850 | 340k | ■ 5.20 | ■ 4.15 | **+■ 284.1** | **12.4%** | ■ ACCEPT |
| Khambhat Offshore Wind | Wind | ■ 920 | 260k | ■ 6.40 | ■ 5.80 | **+■ 82.5** | **10.8%** | ■ ACCEPT |

## 4. Final Course Rubric Compliance & Conclusion

• **Complete Pipeline:** Ingests Kaggle World Bank macroeconomic series, cleans them into project-level empirical benchmarks, and runs multi-stage financial valuations.
• **Real Decision Value:** Distinguishes viable commercial assets from value-destructive contracts, tests banking solvency (DSCR), and optimizes renewable energy capital mix.
• **Cloud Deployed Application:** https://capital-budgeting-project-karthickruban.streamlit.app